



Credit Card Fraud Detection: Train and Evaluate Machine Learning Classifier Models

1. Executive Summary

This report presents the analysis and evaluation of three distinctive machine learning classifier models to detect fraudulent credit card transactions for a fraud detection system:

1. Logistic Regression (baseline model)

2. Random Forest

3. XGBoost

The dataset selected for this report is [the credit card fraud detection dataset from Kaggle.](#)

*With the increasing prevalence of digital transactions, robust fraud detection systems are crucial for financial institutions to minimize losses and maintain customer trust. I aim to compare, select, and train a classifier model that effectively identifies fraudulent activities while minimizing false positives. My approach combines techniques such as the **Synthetic Minority Oversampling Technique (SMOTE)** with practical business insights to enhance financial security and operational efficiency.*

2. Main Objectives

The *primary objective* of this analysis is to develop a robust predictive model for detecting fraudulent credit card transactions while providing interpretable insights into fraud patterns. Our focus is dual:

1. Prediction: Create a high-accuracy model to identify potential fraudulent transactions in real-time.
2. Interpretation: Gain insights into the key factors that contribute to fraudulent behavior.

Benefits to stakeholders:

- Financial: Significant reduction in losses from fraudulent activities
- Customer Trust: Enhanced fraud protection leading to improved customer satisfaction and loyalty.
- Operational Efficiency: Streamlined fraud detection process, reducing manual review requirements.
- Strategic Insights: Data-driven understanding of fraud patterns to inform preventive measures.

Outline and goals for this analysis

1. Develop a model that accurately identifies fraudulent transactions despite severe class imbalance.
2. Identify the most important features for fraud detection.
3. Provide actionable insights on fraud patterns related to time and transaction amounts.

4. Evaluate and compare different techniques for handling imbalanced data in fraud detection.
5. Recommend a model based on the analysis and evaluation previously performed
6. Share the key findings and insights
7. Discuss the model limitations and next steps

3. Data Description

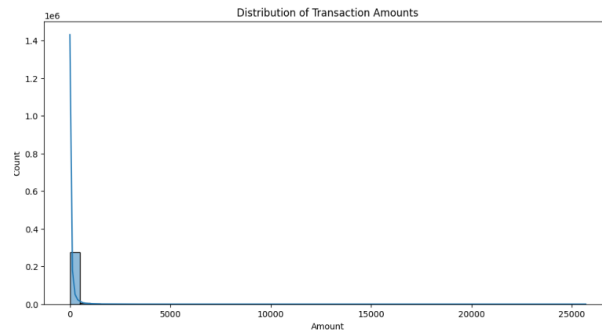


4.1 Exploratory Data Analysis (EDA)

EDA Summary:

Transaction Amounts

1. The majority of transactions are of small amounts, but there are some high-value outliers.
2. Fraudulent transactions tend to have lower amounts on average compared to non-fraudulent ones



1 - Figure 2: Distribution of Transaction Amounts

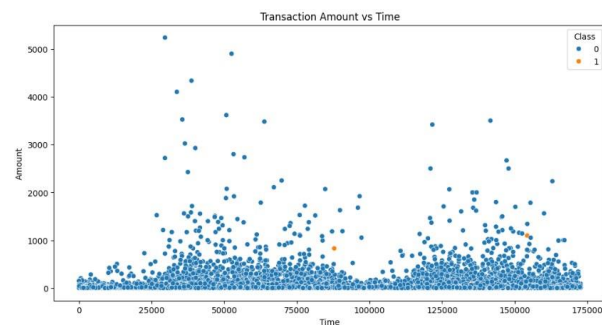
- *The majority of transactions are of small amounts, but there are some high-value outliers.*



2 - Figure 3: Transaction Amounts by Class

Time Analysis

- Transactions occur over a period of about two days (172,792 seconds).
- There's no clear pattern in the relationship between time and transaction amount.
- Certain hours of the day show higher frequencies of fraudulent transactions.

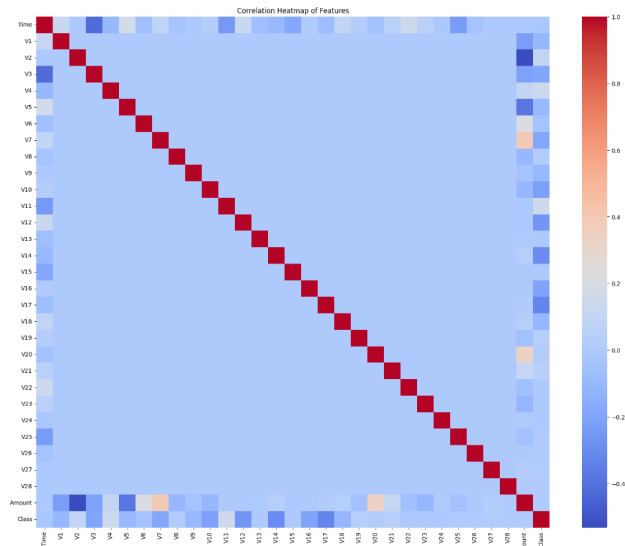


3 - Figure 4: Transaction Amount vs Time

Feature Correlations

- Some anonymized features (e.g., V14, V4, V10) show strong negative correlations with the Class variable, indicating they might be good predictors of fraud.

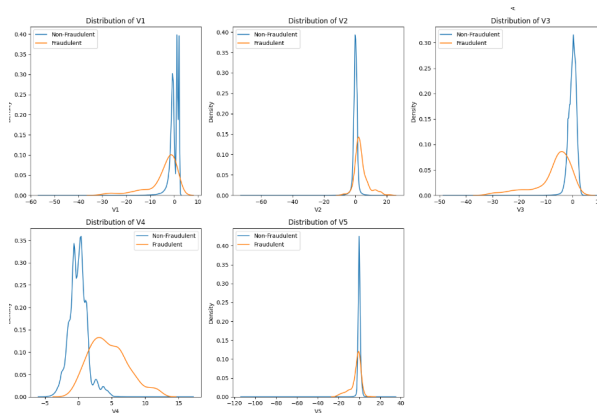
- There are correlations between some of the anonymized features, suggesting potential multicollinearity.



4 - Figure 5: Correlation Heatmap of Features

Anonymized Features

- Several anonymized features (V1-V28) show distinct distributions for fraudulent and non-fraudulent transactions, particularly V1-V5.



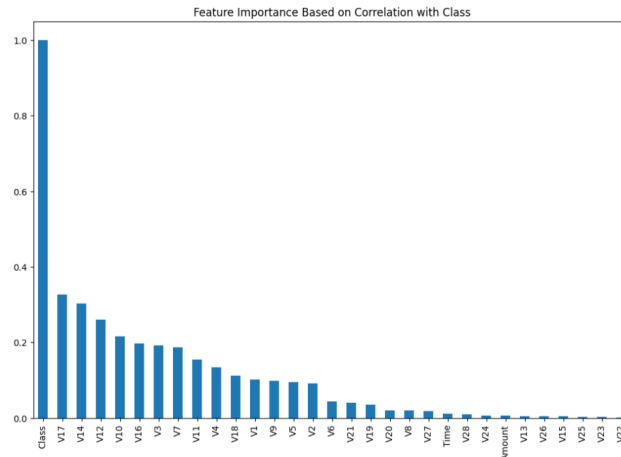
5 - Figure 6: Distribution of anonymized features

Feature Importance

- Based on correlation with the Class variable, features like V14, V4, V10, V12, and V17 appear to be the most important for detecting fraud.

Feature	Correlation
Class	1.000000
V17	0.326481
V14	0.302544
V12	0.260593
V10	0.216883
V16	0.196539
V3	0.192961
V7	0.187257
V11	0.154876
V4	0.133447
V18	0.111485
V1	0.101347
V9	0.097733
V5	0.094974
V2	0.091289
V6	0.043643
V21	0.040413
V19	0.034783
V20	0.020090
V8	0.019875
V27	0.017580
Time	0.012323
V28	0.009536
V24	0.007221
Amount	0.005632
V13	0.004570
V26	0.004455
V15	0.004223
V25	0.003308
V23	0.002685
V22	0.000805

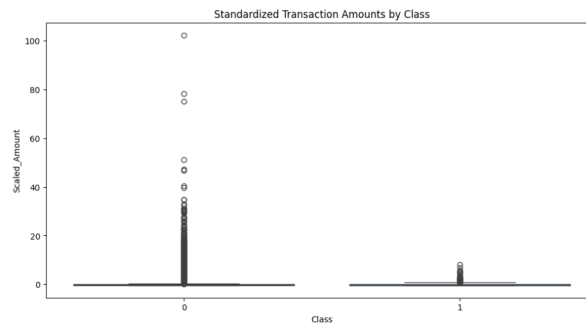
Note: Data type is float64



7 - Figure 8: Feature Importance Bar Chart Based on Correlation with Class

Standardized Amounts

- Even after standardization, fraudulent transactions show a different distribution of amounts compared to non-fraudulent ones.



8 - Figure 9: Standardized Transaction Amount by Class

Key Insights from EDA

- The extreme class imbalance (0.17% fraud) will require careful handling in the modeling phase.
- Transaction amount and time of day could be useful features for fraud detection.
- Several of the anonymized features show promise as strong predictors of fraud.
- The differences in feature distributions between fraudulent and non-fraudulent transactions suggest that machine learning models could be effective in distinguishing between the two classes.